



Q1 [50 points]: A fixed-diameter cylindrically symmetric (volumetric) charge configuration extends to infinity in the $\pm z$ directions; the radius of the cylinder is $\rho = a$. Inside the radius b , where $b < a$, the charge density has a parabolic dependence on radius while over the range $b < \rho < a$ outside that radius, the volumetric charge density is zero.

$$\rho_v = \begin{cases} \rho_o \frac{\rho^2}{b^2} & \rho < b \\ 0 & b < \rho < a \end{cases}$$

(a) [40 points] Use the axial symmetry and Gauss' integral law to show that $\mathbf{E}$ in the two regions is

$$\mathbf{E} = \begin{cases} \frac{\rho_o \rho^3}{4\epsilon_o b^2} \mathbf{a}_\rho & \rho < b \\ \frac{\rho_o b^2}{4\epsilon_o \rho} \mathbf{a}_\rho & b < \rho < a \end{cases}$$

(b) [10 points] Outside the cylinder at $\rho > a$, the electric field is zero, $\mathbf{E} = 0$. This is made possible by implanting deliberately a surface charge density on the outer surface of the cylinder. Using the boundary condition, $\epsilon_1 \mathbf{E}_1 - \epsilon_2 \mathbf{E}_2 = \rho_s \mathbf{a}_n$, obtain the surface charge density at $\rho = a$.

Note: Since, no dielectric material is specified here, you may assume that $\epsilon_1 = \epsilon_2 = \epsilon_0$, where ϵ_0 is the permittivity of the free space.

Q2 [20 points]: A cylindrical slab has a polarization given by $\mathbf{P} = \rho_o \sqrt{\rho} \mathbf{a}_\rho$. Find the polarization volumetric charge density ρ_{pv} inside the slab and its polarization surface charge density ρ_{ps} . Note: $\rho_{ps} = \mathbf{P} \cdot \mathbf{a}_n$ and $\rho_{pv} = -\nabla \cdot \mathbf{P}$.

Q3 [30 points]: Point charges $Q_1 = 2\text{nC}$, $Q_2 = -3\text{nC}$, $Q_3 = 4\text{nC}$, and $Q_4 = -2\text{nC}$ are positioned one at a time and in that order at $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, -1)$, and $(0, 0, 1)$, respectively. Calculate the energy in the system after each charge is positioned.

Q4 [10 points]: Inside the cylinder of rectangular cross-section, the potential is given as

$$V = \frac{\rho_o}{\epsilon_o} \cos \frac{\pi}{a} x \cos \frac{\pi}{b} y$$

Find $\mathbf{E}$.

Q1. $\rho_v = \begin{cases} \rho_0 \frac{r^2}{b^2}, & 0 < r < b \\ 0, & b < r < a \end{cases}$

SOLUTION:

(a)

Let us consider a cylindrical gauss's surface of radius $r < b$

According to Gauss's theorem.

$$\oint_S \epsilon_0 \vec{E} \cdot d\vec{S} = Q_{enc.}$$

$\vec{E} = E_r \hat{r}$ due to symmetry.

$$d\vec{S} = r d\phi dz \hat{r}$$

Let the length of Gauss's cylinder be $z = L$.

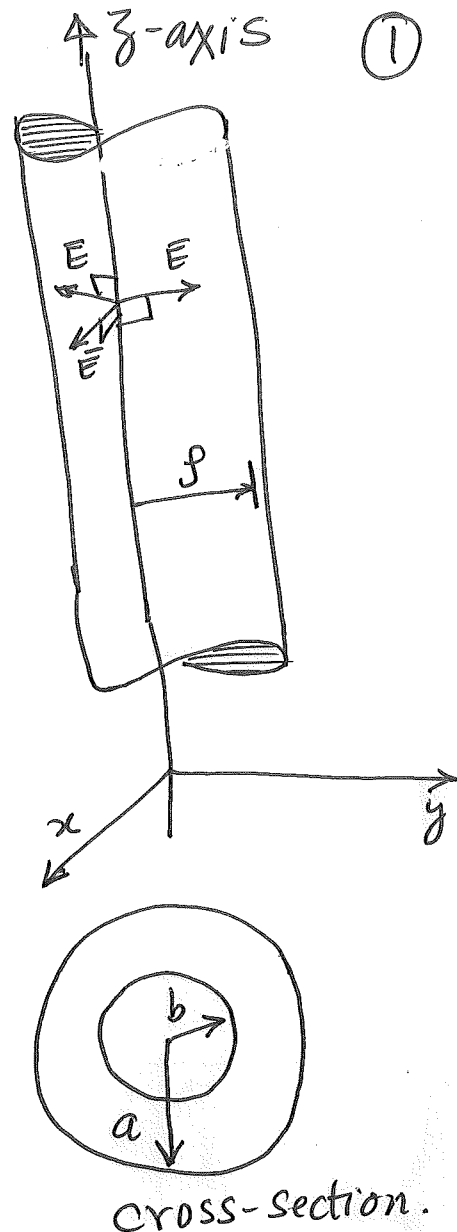
$$L.H.S. = \oint_S \epsilon_0 \vec{E} \cdot d\vec{S}$$

$$= \int_{z=0}^L \int_{\phi=0}^{2\pi} \epsilon_0 E_r \hat{r} \cdot r d\phi dz \hat{r}$$

$$= \epsilon_0 r E_r \int_0^L dz \int_0^{2\pi} d\phi = 2\pi L \epsilon_0 r E_r \quad \text{--- (1)}$$

$$R.H.S. = Q_{enc} = \int_V \rho_v dv = \int_{z=0}^L \int_{\phi=0}^{2\pi} \int_{r'=0}^r \rho_0 \frac{r'^2}{b^2} r' dr' d\phi dz$$

$$= \frac{\rho_0}{b^2} \frac{r'^4}{4} \Big|_0^r \int_0^L dz \int_0^{2\pi} d\phi = \frac{\rho_0 r^4 2\pi L}{4 b^2} \quad \text{--- (2)}$$



Equating ① & ②, we get.

②

$$2\pi L \epsilon_0 \rho E_\rho = \frac{\rho_0 \rho^3 2\pi L}{4b^2}$$

$$\Rightarrow E_\rho = \frac{\rho_0 \rho^3}{4\epsilon_0 b^2}$$

$$\text{or } \boxed{\vec{E} = \frac{\rho_0 \rho^3}{4\epsilon_0 b^2} \hat{a}_\rho}$$

for $0 < \rho < b$

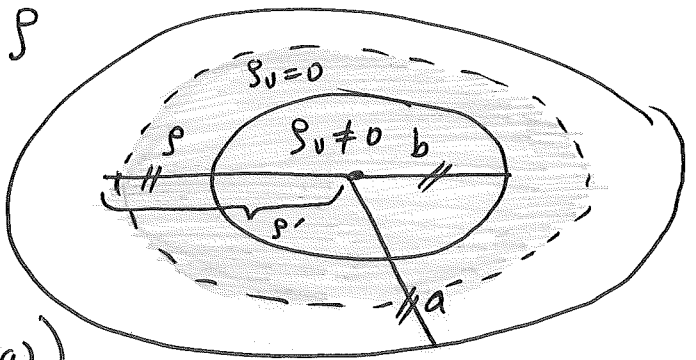
PROVED.

Now consider $b < \rho < a$
and $0 < \rho' < \rho$

Again

$$L.H.S. = 2\pi\epsilon_0 L \rho E_\rho \quad \text{--- (3)}$$

(as we obtained in part (a))



$$\begin{aligned} R.H.S. &= Q_{enc} = \int_V \rho_v dv = \int_{z=0}^L \int_{\phi=0}^{2\pi} \int_{\rho=0}^a \rho_v dv \\ &= \int_{z=0}^L \int_{\phi=0}^{2\pi} \int_{\rho=0}^b \frac{\rho_0 \rho^2}{b} \rho d\rho d\phi dz \\ &= \frac{\rho_0}{4} b^2 2\pi L. \quad \text{--- (4)} \end{aligned}$$

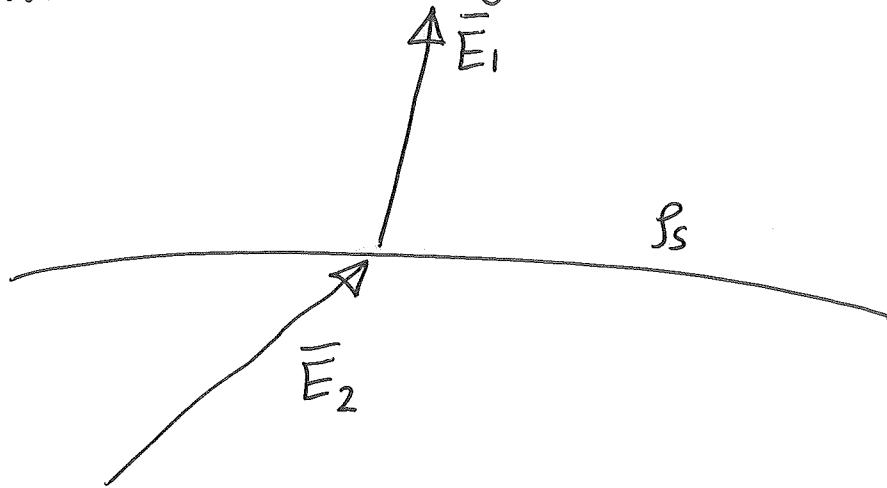
Equating ③ & ④, we obtain.

$$\boxed{\vec{E} = \frac{\rho_0 b^2 \hat{a}_\rho}{4\epsilon_0 \rho}}$$

for $\rho > b$

(b) It is given that $\vec{E} = 0$ outside the cylinder ⁽³⁾
at $\rho \geq a$.

Recall the boundary condition theory.



We have $D_{1n} - D_{2n} = \rho_s$
or $\overline{D}_{1n} - \overline{D}_{2n} = \rho_s a_n$ — (5)

In our problem, just before $\rho \leq a$, we have

$$\vec{E}_2 = \frac{\rho_0 b^2}{4\epsilon_0 \rho} a_\rho \Big|_{\rho=a} = \frac{\rho_0 b^2}{4a\epsilon_0} a_\rho.$$

just after $\rho \geq a$, we have

$$\vec{E}_1 = 0$$

From (5), we get $(a_n = a_\rho)$.
 $\epsilon_1 E_1 - \epsilon_2 E_2 = \rho_s$

$$\boxed{\rho_s = -\epsilon_0 E_2 \Big|_{\rho=a} = -\frac{\rho_0 b^2}{4a}}$$

Q2. SOLUTION:

④

$$\bar{P} = \epsilon_0 \sqrt{\epsilon} \epsilon_f$$

Polarization
Surface
charge
Density

$$= \rho_{ps} = \bar{P} \cdot \hat{a}_n$$

$$= \bar{P} \cdot \hat{a}_f$$

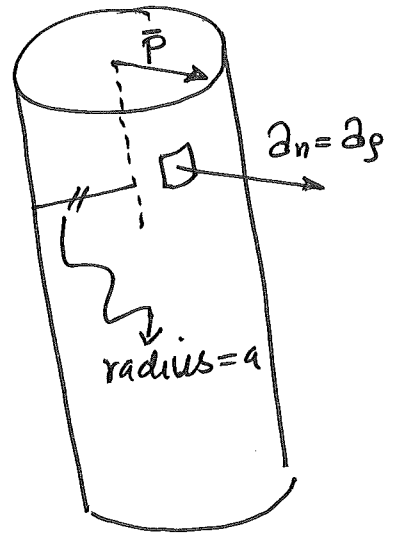
$$= \epsilon_0 \sqrt{\epsilon} \epsilon_f \cdot \hat{a}_f$$

(at surface)

$$\rho_{ps} = \epsilon_0 \sqrt{\epsilon} \epsilon_f \text{ (at surface)}$$

Let the radius be $\epsilon = a$.

$$\boxed{\rho_{ps} = \epsilon_0 \sqrt{a}}$$



Polarization
Volumetric
charge
Density.

$$= \rho_{pv} = -\nabla \cdot \bar{P}$$

$$= -\frac{1}{\epsilon} \frac{\partial}{\partial \epsilon} (\epsilon \epsilon_0 \sqrt{\epsilon})$$

$$= -\frac{\epsilon_0}{\epsilon} \frac{\partial}{\partial \epsilon} (\epsilon^{3/2})$$

$$= -\frac{\epsilon_0}{\epsilon} \frac{3}{2} \epsilon^{3/2-1}$$

$$\boxed{\rho_{pv} = \frac{3}{2} \frac{\epsilon_0 \sqrt{\epsilon}}{\epsilon} = -\frac{3}{2} \epsilon_0 \frac{1}{\sqrt{\epsilon}}}$$

Q3.

$$Q_1 = 2nC \text{ @ } (1, 0, 0)$$

$$Q_2 = -3nC \text{ @ } (0, 1, 0)$$

$$Q_3 = 4nC \text{ @ } (0, 0, -1)$$

$$Q_4 = -2nC \text{ @ } (0, 0, 1)$$

Let us bring these charges in order 1, 2, 3, & 4.

$$W_E = Q_1 V_1 + Q_2 V_{21} + Q_3 V_{31} + Q_4 V_{41} + Q_3 V_{32} + Q_4 V_{42} + Q_4 V_{43}$$

We bring Q_1 in ^{an empty} (vacuum) region having no field.

So $V_1 = 0$. $W_1 = Q_1 V_{21} = 0$

We bring Q_2 ,

$$V_{21} = \frac{Q_1}{4\pi\epsilon_0 |(1, 0, 0) - (0, 1, 0)|} = \frac{Q_1}{4\pi\epsilon_0 \sqrt{2}}$$

$$W_2 = Q_2 V_{21} = \frac{Q_1 Q_2}{4\pi\epsilon_0 \sqrt{2}}$$

We bring Q_3 .

$$V_{31} = \frac{Q_1}{4\pi\epsilon_0 |(1, 0, 0) - (0, 0, -1)|} = \frac{Q_1}{4\sqrt{2}\pi\epsilon_0}$$

$$V_{32} = \frac{Q_2}{4\pi\epsilon_0 |(0, 1, 0) - (0, 0, -1)|} = \frac{Q_2}{4\sqrt{2}\pi\epsilon_0}$$

$$W_3 = Q_3 (V_{31} + V_{32}) = \frac{Q_3 (Q_1 + Q_2)}{4\sqrt{2}\pi\epsilon_0}$$

We bring Q_4 .

(6)

$$V_{41} = \frac{Q_1}{4\pi\epsilon_0 |(1, 0, 0) - (0, 0, 1)|} = \frac{Q_1}{4\sqrt{2}\pi\epsilon_0}$$

$$V_{42} = \frac{Q_2}{4\pi\epsilon_0 |(1, 0, 0) - (0, 0, 1)|} = \frac{Q_2}{4\sqrt{2}\pi\epsilon_0}$$

$$V_{43} = \frac{Q_3}{4\pi\epsilon_0 |(0, 0, -1) - (0, 0, 1)|} = \frac{Q_3}{8\pi\epsilon_0}$$

$$\begin{aligned} W_4 &= Q_4 (V_{41} + V_{42} + V_{43}) \\ &= \frac{Q_4(Q_1 + Q_2)}{4\sqrt{2}\pi\epsilon_0} + \frac{Q_4 Q_3}{8\pi\epsilon_0} \end{aligned}$$

Finally.

$$W_E = W_1 + W_2 + W_3 + W_4.$$

Q4 SOLUTION.

$$V = \frac{\rho_0}{\epsilon_0} \cos\left(\frac{\pi x}{a}\right) \cos\left(\frac{\pi y}{b}\right)$$

$$\vec{E} = -\nabla V = -\frac{\partial V}{\partial x} \hat{a}_x - \frac{\partial V}{\partial y} \hat{a}_y$$

~~$$\vec{E} = \frac{\pi \rho_0}{a \epsilon_0} \sin\left(\frac{\pi x}{a}\right) \hat{a}_x + \frac{\pi \rho_0}{b \epsilon_0} \sin\left(\frac{\pi y}{b}\right) \hat{a}_y.$$~~

$$\vec{E} = \frac{\pi \rho_0}{a \epsilon_0} \sin\left(\frac{\pi x}{a}\right) \cos\left(\frac{\pi y}{b}\right) \hat{a}_x$$

$$+ \frac{\pi \rho_0}{b \epsilon_0} \sin\left(\frac{\pi y}{b}\right) \cos\left(\frac{\pi x}{a}\right) \hat{a}_y.$$